RAHUL MITRA

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EDUCATION

National University of Singapore **Aug 2023 – Jul 2027**
Bachelor of Engineering, Industrial and Systems Engineering (GPA: 4.37/5.00)

- Pursuing a Second Major in Computer Science and a Minor in Mathematics.
- Awarded Certificate of Outstanding Performance in 3D Computer Vision (CS4277): top student in a class of 24, NUS School of Computing.

WORK EXPERIENCE

Warehouse AI & Operations Research Engineer (Part-Time), STMicroelectronics (NUS Capstone) **Aug 2026 – Present**

- Architected a layout-agnostic AI put-away engine ingesting each site's shelf coordinates, dimensions, and operating rules, letting one model adapt to any STMicroelectronics warehouse floor plan without per-site rework.

Global Data Engineer (Operational AI Systems), Part-Time, Abbott Laboratories (Contract) **Jun 2026 – Present**

- Developed ADAM, an offline tool-calling assistant on a locally hosted Gemma 4 E4B model (llama.cpp) answering spray-dryer setpoint, process, and signal questions for 6 dryers across 4 global sites, giving engineers self-serve answers to decisions previously requiring APC specialists, with rollout to Abbott servers scheduled.
- Converted 4 sites' Advanced Process Control master files from Excel into 7 steady-state gain models solved as HiGHS linear programs and SLSQP tracking QPs, returning setpoints, binding constraints, and shadow prices in milliseconds across 5 modes: APC replica, single-setpoint push, CV targeting, what-if limit reports, and operator what-ifs.

Data Analytics & Product Management Intern, Abbott Laboratories (Converted to Contract) **Jan 2026 – Jun 2026**

- Developed a SimPy discrete-event digital twin and OR-Tools CP-SAT solver for hybrid flow-shop scheduling, then compared 3 approaches (heuristic, MIP, and genetic algorithm) to turn production objectives and constraints into decision-ready schedules.
- Validated a non-destructive, 15-stage changeover-data pipeline against 5 years of unseen raw data with zero execution failures across operator- and machine-generated anomalies.

Software Engineer (Full Stack & Cloud), People's Association (Pro bono) **Sep 2025 – Sep 2026**

- Recruited into PA's Sparks Citizens Developer network as a community innovator in recognition of prior public-good initiatives, then designed, built, and deployed Churp, a resident-driven garden-plot balloting platform now live in Tengah and slated for national rollout, seeded by a S$20K Sparks Community Innovation Fund award.

RESEARCH EXPERIENCE

Computer Vision Researcher, BlendED x Amazon Robotics (AI+X Global Challenge) **Jul 2026 – Oct 2026**

- Built vision systems turning raw camera feeds into perception-ready frames across 3 workstreams: camera ISP, super-resolution, and image restoration under real-world blur.

PROJECTS

Team Lead & Finalist, Swarmline: SG Defence Tech Hackathon (1,300+ applicants) **Sep 2026**

- Directed a 5-person team to architect decentralised drone-swarm coordination on correlated random-walk search, gradient-flow dispersion, delay-tolerant gossip, cross-inhibition consensus, and response-threshold task allocation.
- Implemented a deterministic Unreal Engine 5.8 C++ simulation with Claude Code; under a ground-link jam, 30 drones confirmed all 8 walking targets in 10 of 10 seeded runs, versus 2.1 of 8 on average for an operator-run fleet.

Finalist, Volt Pulse SG: SMU Hack For Cities (Top 8 Finalist of 50+ teams) **Jan 2026**

- Integrated SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning into an autonomous loop managing electricity and utility savings across multiple HDB households at once.

Team Lead, Maritime Deficiency Severity Forecasting: Maritime Hackathon **Jun 2025**

- Led model training for a maritime-inspection NLP system, fine-tuning a 109M-parameter BERT with DNN layers and gradient clipping on the ASPIRE 2A supercomputer.

CO-CURRICULAR ACTIVITIES AND VOLUNTEER EXPERIENCES

Elderly Care Volunteer, NTUC Health Active Ageing Centre **Sep 2025 – Sep 2026**

- Coordinated weekly senior-engagement activities, including home visits to check on residents' wellbeing, hear life stories, and help with everyday tasks, building empathy for accessible, community-focused product design.

NATIONAL SERVICE

Republic of Singapore Navy, Naval Chef & Base Support Assistant **Jan 2021 – Jan 2023**

- Compressed a month-long manual analysis of 6 months of food-waste data into a 3-minute Excel VBA reporting workflow inside a security-constrained environment.

SKILLS AND CERTIFICATIONS

Technical: CP-SAT, HiGHS linear programming, SLSQP quadratic programming, SciPy, hybrid flow-shop scheduling, SimPy, digital-twin simulation, robust optimization, graph optimization, network flow, Dijkstra/SSSP/APSP, decision analytics, statistics, AI tools (Claude Code, ChatGPT, Codex, Gemini, ComfyUI), Python, Excel VBA
Certifications: NVIDIA Disaster Risk Monitoring with Satellite Imagery